

Pakshal Bohra

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Summary

Machine Learning Research Engineer with a PhD from EPFL, specializing in generative modeling, representation learning, and computer vision. Proven track record of algorithmic innovation with 7 first-author peer-reviewed publications. Industry experience building multimodal ML systems end-to-end, from research to deployment on edge hardware.

Experience

ML Research Engineer, Distran AG – Zurich, Switzerland Jan 2026 – Present

- Leading the development of a real-time audio-visual ML system for scene understanding on ultrasonic cameras, spanning dataset preparation and foundation-model fine-tuning with contrastive learning
- Optimizing the inference pipeline to deploy trained models under latency and memory constraints on production hardware

Computational Software Engineer, Distran AG – Zurich, Switzerland Aug 2024 – Dec 2025

- Co-developed DeepRange, an acoustic imaging algorithm shipped on Distran's cameras that enables detection of gas leaks 5x fainter than previously possible in typical industrial environments
- Designed and implemented DVC-backed ML pipelines for the full development cycle of interpretable leak quantification models, enabling reproducible experimentation and deployment readiness
- Built a production-grade internal semantic search engine based on vision-language models for text-based image retrieval, replacing manual search workflows
- Core contributor of high-performance signal processing and acoustic imaging pipelines to an internal JAX-based scientific computing SDK for rapid algorithm prototyping and large-scale data reprocessing

Research Assistant, EPFL – Lausanne, Switzerland Sep 2018 – Jun 2024

* denotes first author publications

- Developed a Bayesian physics-guidance algorithm for deep generative models to solve nonlinear imaging inverse problems. Enabled high-quality reconstructions with uncertainty quantification in severely ill-posed conditions [IEEE TCI 2022*]
- Built a deep video prior with a shared image decoder and manifold-structured temporal latent representation for dynamic quantitative phase imaging. Achieved state-of-the-art reconstructions from a single measurement per frame, with up to 10 dB improvement over baselines [Inv. Problems 2023*]
- Introduced a scalable PyTorch framework for the efficient learning of activation functions in deep neural networks. Leveraged this framework to construct highly competitive Lipschitz-constrained networks for robust learning [IEEE OJSP 2020*, NeurIPS Workshop 2021*, JMLR 2024]
- Built a statistical benchmarking toolkit for deep-learning-based solutions to inverse problems, using MCMC posterior samplers as gold-standard references for optimality evaluation [IEEE TSP 2023*]
- Proposed a neural-network-based convex regularizer for fast and interpretable image reconstruction, and applied it to MRI and CT imaging. Awarded the 2026 SIAM Activity Group on Imaging Science Best Paper Prize [IEEE TCI 2023]
- Published 11 peer-reviewed papers (7 as first author); presented technical tutorials at SIAM-IS and IEEE ISBI on Bayesian Inference for Inverse Problems
- Supervised 13 student research projects (master's theses, internships, and semester projects); served as Teaching Assistant for Image Processing I and II

Research Intern, Siemens Healthineers – Bengaluru, India

May 2016 – Jul 2016

- Developed an algorithm based on supervoxel features, random forest classifiers and the graph cuts method for the automated segmentation of the spleen in CT images

Personal Projects

Flow Matching with Diffusion Transformers [[Code](#)] [[Blog](#)]

Mar 2026 – Present

- Implemented flow matching for image generation from scratch in pixel and latent spaces with the DiT architecture. Extended to text-conditioned generation using MM-DiT with classifier-free guidance. Focused on training efficiency under single-GPU compute and memory constraints

Education

PhD in Electrical Engineering, EPFL – Lausanne, Switzerland

Sep 2018 – Jun 2024

- Advisor: Prof. Michael Unser
- Research Areas: Deep Learning, Generative Modeling, Inverse Problems, Statistical Inference

B.Tech + M.Tech in Electrical Engineering, IIT Bombay – Mumbai, India

Jul 2013 – Jun 2018

- Minor in Computer Science and Engineering

Technical Skills

AI/ML Topics: Flow Matching and Diffusion Models, Representation Learning, Multimodal Learning, Language Models, Computer Vision, Transformers

Programming Languages: Python, C++ , Java

Frameworks and Libraries: PyTorch, JAX, Equinox, Hugging Face (Transformers, Diffusers, PEFT, Accelerate), TensorRT, ONNX, XGBoost, scikit-learn, NumPy, SciPy, pandas, polars, OpenCV, Streamlit

Dev/ML Ops & Infrastructure: Google Cloud Platform (GCP), Git, DVC, MLflow, Docker, Jenkins

Selected Publications

† denotes equal contribution

- **P. Bohra**[†], J. Campos[†], H. Gupta, S. Aziznejad, and M. Unser, “[Learning Activation Functions in Deep \(Spline\) Neural Networks](#)”, *IEEE Open Journal of Signal Processing*, 2020
- **P. Bohra**, T. -a. Pham, J. Dong, and M. Unser, “[Bayesian Inversion for Nonlinear Imaging Models Using Deep Generative Priors](#)”, *IEEE Transactions on Computational Imaging*, 2022
- **P. Bohra**[†], T. -a. Pham[†], Y. Long, J. Yoo, and M. Unser, “[Dynamic Fourier Ptychography With Deep Spatiotemporal Priors](#)”, *Inverse Problems*, 2023
- **P. Bohra**, D. Perdios, A. Goujon, S. Emery, and M. Unser, “[Learning Lipschitz-Controlled Activation Functions in Neural Networks for Plug-and-Play Image Reconstruction Methods](#)”, *NeurIPS Workshop on Deep Learning and Inverse Problems*, 2021
- **P. Bohra**, P. del Aguila Pla, J. -F. Giovannelli, and M. Unser, “[A Statistical Framework To Investigate the Optimality of Signal-Reconstruction Methods](#)”, *IEEE Transactions on Signal Processing*, 2023
- A. Goujon, S. Neumayer, **P. Bohra**, S. Ducotterd, and M. Unser, “[A Neural-Network-Based Convex Regularizer for Inverse Problems](#)”, *IEEE Transactions on Computational Imaging*, 2023
- S. Ducotterd, A. Goujon, **P. Bohra**, D. Perdios, S. Neumayer, and M. Unser, “[Improving Lipschitz-Constrained Neural Networks by Learning Activation Functions](#)”, *Journal of Machine Learning Research*, 2024
- R. Parhi, **P. Bohra**, A. El Biari, M. Pourya, and M. Unser, “[Random ReLU Neural Networks as Non-Gaussian Processes](#)”, *Journal of Machine Learning Research*, 2025